

This project is on the sentimental analysis of comments and ratings on the following movies combined together . The movies are ***Gladiator, Jack and Jil, You People , Sardar Udham, Thor: Love and Thunder and Kireedam***. From the movies, a data frame of dataset of the movies review and rating was formed named “movie”

Calling the nltk Library a total of 11,210 words was derived from the review to build a word cloud. However the word gotten is not a total of the 11210 words because some irrelevant word like “movie”, “Made”, “make”, “Im”, “man”, etc were remove from the total “text” to get the important word that portrays sentiment.

Furthermore a machine learning algorithm **called Vader** was used to classify the reviews into **Positive , Negative and Neutral Reviews** . As a result, out of the total of 280 reviews 144 reviews were positive, 110 reviews were negative while 18 was neutral as classified by the **Vader Algorithm** ..

The algorithm classified a lot of the reviews correctly while others it classified them incorrectly .

These are some of this review statement below of which the algorithm classified correctly,

1. Are you entertained? Yes sir yes I am
2. very very disappointed
3. Highly disappointing

From the above reviews statement, number “1” statement is positive and the algorithm classified it as positive while “2” and “3” is negative and the algorithm classified them as negative.

However there are quite a lot the Vader Algorithm did not classified correctly like the review statement below,

1. Below Average
2. Old, dated, cheap
3. Deserves to be Erased from Film History
4. A CLASSIC MOVIE.
5. Who wrote this garbage?

6. It's a joke.

From the above reviews statement number “1, 2, 3, and 5, “are negative reviews in nature but the algorithm classified them as all neutral while number “4” is positive but the algorithm classified it as neutral. Number “6” review is negative in nature but the algorithm classified them as all positive.

After considering the above analysis, using more data or large data set will help with better classification.

Furthermore, two new set of Dataframe was form call “**textfeatures**” and “**textfeatures2**”. “**textfeatures**” is dataframe with neutral classification and “**textfeatures2**” is dataframe without neutral classification.

“**textfeatures2**” is process using the “**countvectorizer.fit_transform**” to get a dataframe matrix called “**count_matrix**” and “**tfidfvectorizer.fit_transform**” process to get a dataframe matrix called “**tfidf_matrix**” and the result is that the dimension of **count_matrix** display X rows (262 reviews), and Y columns (314 word features) in the transformed dataset.

When “**textfeatures2**” without pre-processing is analyzed the dimension of **count_matrix** are X rows (262 reviews), and Y columns (391 word features) in the transformed dataset

in addition a lot of models are built in the worksheet, the first models built with the “**tfidf_matrix**” scheme using the **DecisionTree model** and the **SVC model**.

Both models is built using 70percent of the whole data to train the model with a random state of 94.

The result from both model are with very high accuracy but the SVC model has a higher **Accuracy** of 94 with **Precision** of 95 , while **DecisionTree Model** has an **Accuracy** of 92 with a **Precision** of 92.

Also noticed is that in both models, the machine predicted the negative review with a high **Accuracy** more than it can predict the positive review.

The second model built is with the “**count_matrix**” using the the **DecisionTree model and SVC model**.

Both model are also built using 70percent of the whole data to train the model with a random state of 94.

The result from both model are with very high **Accuracy** but this time with “count_matrix “ the DecisionTree has a higher **Accuracy** of 92 with **Precision** of 93 , while the SVC model has an **Accuracy** of 83 with a **Precision** of 83.

Also noticed is that in both models, the machine predicted the positive review accurately more than it can predict the negative review.

REFERENCE

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